MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, FLEXIBLE, TWISTABLE, CLASS 1

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-287

REQUIREMENTS See table I

Part number See table I

Extent of qualification Qualification obtained for one assembly shall constitute qualification for other assemblies as specified in table II

Cross reference information See table III

Substitution data See table IV

FSC 5985

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TABLE I Flexi

Group number	Part number M28[illegible] 3-	TE_{11} mode frequency range (GHz	Minimum centerline bending radius (inch)		Repeated twist degrees/ft		Insertion loss (dB/ft)	Axial twist (degrees/ft)	Pressurization lb_f/in^2 gage	$\frac{2}{\lambda}$ VS[illegible] (λ_{ax}) at	
			E plane	H plane	400 cycles	$\times 10^5$ cycles				.2	25
I	01-XXXX 02-XXXX 60-XXXX	1 12-1 70	24 12	48 08	5 0	1 5	0 005	10	2	1 05	1 06
	03-XXXX 04-XXXX 61-XXXX	1 70-2 60	12 39	25 95	7 50	2 25	0 01	15	4	1 06	1 07
	05-XXXX 06-XXXX 62-XXXX	2 20-3 30	11 14	19 64	11 0	3 3	0 015	22	5	1 06	1 07
	07-XXXX 08-XXXX 09-XXXX 10-XXXX 11-XXXX 12-XXXX 13-XXXX 14-XXXX 63-XXXX	2 60 3 95	6 88	14 38	14 5	4 5	0 02	28	15	1 08	1 09
II	15-XXXX 16-XXXX 17-XXXX 18-XXXX 19-XXXX 20-XXXX 21-XXXX 22-XXXX 64-XXXX	3 95-5 85	4 94	12 04	25 0	7 5	0 05	50	30	1 06	1 09
	23-XXXX 24-XXXX 25-XXXX 26-XXXX 27-XXXX 28-XXXX 29-XXXX 30-XXXX 65-XXXX	5 85-8 20	3 88	9 25	35 0	10 5	0 08	70	30	1 03	1 09
	31-XXXX 32-XXXX 33-XXXX 34-XXXX 35-XXXX 36-XXXX 37-XXXX 38-XXXX 66-XXXX	7 05-10 0	3 57	8 38	42 5	12 75	0 08	85	30	1 12	1 14

See footnotes at end of table

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waveguide assemblies

Flexure (degrees/ft), 1x10^5 cycles		Jacket material	Retention force (lbf)	Oper temp range -55 to	Mates with standard rigid waveguide MBS/1-	Flanges		
E plane	H plane					Flange 1 M3922/'	Flange 2 M3922 ( )	Material 3/
4 5 4 5 4/	2 2 2 2 4/	Neoprene	N/A	+100°C	017 018 017	58-007 (contact) 58-008 (contact) 58-007 (contact)	58-007 (contact) 58-008 (contact) 58-007 (contact)	CU AL CU
7 50 7 50 4/	3 6 3 6 4/	Neoprene	N/A	+100°C	031 029 031	58-009 (contact) 58-010 (contact) 58-009 (contact)	58-009 (contact) 58-010 (contact) 58-009 (contact)	CU AL CU
9 0 9 0 4/	5 25 5 25 4/	Neoprene	N/A	+100°C	037 035 037	58-011 (contact) 58-012 (contact) 58-011 (contact)	58-011 (contact) 58-012 (contact) 58-011 (contact)	CU AL CU
13 3 4/	6 4 4/	Neoprene	N/A	+100°C	043 041 043 041 043 041 043 041 043	64-001 (contact) 64-002 (contact) 56-001 (cover) 56-002 (cover) 61-002 (choke) 61-001 (choke) 61-002 (choke) 61-001 (choke) 61-002 (choke)	64-001 (contact) 64-002 (contact) 56-001 (cover) 56-002 (cover) 56-001 (cover) 56-002 (cover) 61-002 (choke) 61-001 (choke) 56-001 (cover)	CU AL CU AL CU AL CU AL CU
18 7 4/	7 6 4/	Neoprene	N/A	+100°C	055 053 055 053 055 053 055 053 055	63-001 (contact) 63-005 (contact) 57-002 (cover) 57-001 (cover) 62-002 (choke) 62-001 (choke) 62-002 (choke) 62-001 (choke) 63-001 (contact)	63-001 (contact) 63-005 (contact) 57-002 (cover) 57-001 (cover) 57-002 (cover) 57-001 (cover) 62-002 (choke) 62-001 (choke) 63-001 (contact)	CU AL CU AL CU AL CU AL CU
24 0 4/	10 2 4/	Neoprene	N/A	+100°C	067 065 067 065 067 065 067 065 067	63-002 (contact) 63-006 (contact) 55-001 (cover) 55-002 (cover) 60-001 (choke) 60-002 (choke) 60-001 (choke) 60-002 (choke) 63-002 (contact)	63-002 (contact) 63-006 (contact) 55-001 (cover) 55-002 (cover) 55-001 (cover) 55-002 (choke) 60-001 (choke) 60-002 (choke) 63-002 (contact)	CU AL CU AL CU AL CU AL CU
26 1 4/	11 1 4/	Neoprene	N/A	+100°C	073 071 073 071 073 071 073 071 073	63-003 (contact) 63-007 (contact) 53-002 (cover) 53-004 (cover) 59-007 (choke) 59-009 (choke) 59-007 (choke) 59-009 (choke) 63-003 (contact)	63-003 (contact) 63-007 (contact) 53-002 (cover) 53-004 (cover) 53-002 (cover) 53-004 (cover) 59-007 (choke) 59-009 (choke) 63-003 (contact)	CU AL CU AL CU AL CU AL CU

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TABLE I Flexible, twistable

Group number	1/ Part number M28//3-	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Repeated twist (degrees/ft)		Insertion loss (dB/ft)	Axial twist (degrees ft)	Pressurization lb_f/in^2 gage	2/ VSWR (max) at			
			E plane	H plane	400 cycles	1×10^5 cycles				12"	24"	36"	>36
II	39-XXXX 40-XXXX 41-XXXX 42-XXXX 43-XXXX 44-XXXX 45-XXXX 46-XXXX 67-XXXX	8.20-12.40	3.38	7.51	47.5	14.25	0.1	95	30	1.14	1.15	1.16	1.19
III	47-XXXX 48-XXXX 49-XXXX 50-XXXX 51-XXXX 52-XXXX 53-XXXX 68-XXXX	12.4-18.00	2.75	5.75	60.0	18.0	0.25	120	30	1.13	1.15	1.19	1.24
	54-XXXX 55-XXXX 56-XXXX 57-XXXX 58-XXXX 59-XXXX 69-XXXX	18.0-26.5	2.33	4.5	82.5	24.25	0.35	165	30	1.14	1.16	1.20	1.25

1/ The complete part number shall be the number shown and a four-digit number indicating the nominal relaxed length in inches
2/ The VSWR value for any length between the lengths specified shall be the next higher value (e.g. for a 15-inch length of M287/3-01 the VSWR would be 1.06:1)
3/ CU = copper alloy, AL = aluminum alloy
4/ These assemblies are constructed from beryllium copper and the flexure cycle is 10^6 rather than 10^5. All other characteristics are as listed

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Waveguide assemblies - Continued

Flexure (degrees/ft) 1x10^5 cycles		Jacket material	Retention force (lb_f)	Oper temp range -55 to	Mates with standard rigid waveguide MBS/1-	Flanges		
E plane	H plane					Flange 1 M3922/( )	Flange 2 M3922/	Material 3/
27.5	12.3	Neoprene	N/A	+100°C	079	63-004 (contact)	63-004 (contact)	CU
					077	63-008 (contact)	63-008 (contact)	AL
					079	53-001 (cover)	53-001 (cover)	CU
					077	53-003 (cover)	53-003 (cover)	AL
					079	59-006 (choke)	53-001 (cover)	CU
					077	59-008 (choke)	53-003 cover	AL
			100		079	59-006 (choke)	59-006 (choke)	CU
			100		077	59-008 (choke)	59-008 (choke)	AL
4/	4/		N/A		079	63-004 (contact)	63-004 (contact)	CU
33.7	16.2	Neoprene	N/A	+100°C	089	53-005 (cover)	53-005 (cover)	CU
					090	53-006 (cover)	53-006 (cover)	AL
					089	59-001 (choke)	53-005 (cover)	CU
					090	59-002 (choke)	53-006 (cover)	AL
					089	59-001 (choke)	59-001 (choke)	CU
					090	59-002 (choke)	59-002 (choke)	AL
		Plastic neoprene		+65°C	089	53-005 (cover)	53-005 (cover)	CU
4/	4/			+100°C	089	53-005 (cover)	53-005 (cover)	CU
45.0	22.5	Neoprene	N/A	+100°C	102	54-001 (cover)	54-001 (cover	CU
					103	54-002 (cover)	54-002 (cover)	AL
					102	59-003 (choke)	54-001 (cover)	CU
					103	59-004 (choke)	54-002 (cover)	AL
					102	59-003 (choke)	59-003 (choke)	CU
					103	59-004 (choke)	59-004 (choke)	AL
4/	4/				102	54-001 (cover)	54-001 (cover	CU

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TABLE II Grouping for qualification

Group number	Qualifying dash number	Dash numbers of assemblies qualified
I	04	01, 02, 03, 05, 06, 07, 08, 09, 10, 11, 12, 13, and 14
I	61	60, 62, and 63
II	24	15 thru 23, and 25 thru 46
II	65	64, 66 and 67
III	47	48 thru 53
III	55	54, 56 thru 59
III	69	68

TABLE III Cross reference (type designation to part number)

Type designation	Part number M287/3- ( )	Type designation	Part number M287/3- ( )
FG650AEEB	01	FG042ACCB	56
FG430AEEB	03	FG284ACCA	12
FG340AEEB	05	FG187ACCA	20
FG284AEEB	07	FG137ACCA	28
FG187AEEB	15	FG112ACCA	36
FG137AEEB	22	FG090ACCA	44
FG112AEEB	31	FG062ACCA	50
FG090AEEB	39	FG042ACCA	57
FG650AEEA	02	FG284ADDB	09
FG430AEEA	04	FG187ADDB	17
FG340AEEA	06	FG137ADDB	25
FG284AEEA	08	FG112ADDB	33
FG187AEEA	16	FG090ADDB	41
FG137AEEA	24	FG062ADDB	47
FG112AEEA	32	FG042ADDB	54
FG090AEEA	40	FG284ADDA	10
FG284ACCB	11	FG187ADDA	18
FG187ACCB	19	FG137ADDA	26
FG137ACCB	27	FG112ADDA	34
FG112ACCB	35	FG090ADDA	42
FG090ACCB	43	FG062ADDA	48
FG062ACCB	49	FG042ADDA	55

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TABLE IV Substitution data 1/

Dash number	Manufacturer's code (FSCM)	Manufacturer's part number
01	80006	AT-2E-650
01	87864	TL-1C5064
02	80006	AT-6B-650
02	87864	TL-102065
03	80006	AT-2B-430
03	87864	TL-94158
04	80006	AT-6B-430
04	87864	TL-94154
05	80006	AT-2B-340
06	80006	AT-6E-340
07	80006	AT-12B-284
07	87864	TL-86161
08	80006	AT-16B-284
08	87864	TL-86162
09	80006	AT-2B-284
09	87864	TL-86164
10	80006	AT-6B-284
10	87864	TL-86165
11	80006	AT-3B-284
11	87864	TL-86150
12	80006	AT-7B-284
12	87864	TL-86163
13	80006	AT-4B-284
14	80006	AT-8B-284
15	80006	AT-12B-187
15	87864	TL-78173
16	80006	AT-16B-187
16	87864	TL-78174
17	80006	AT-2B-187
17	87864	TL-78175
18	80006	AT-6B-187
18	87864	TL-78176
19	80006	AT-3B-187
19	87864	TL-78167
20	80006	AT-7B-187
20	87864	TL-78172
21	80006	AT-4B-187
22	80006	AT-8B-187
23	80006	AT-12B-137
23	87864	TL-72312
24	80006	AT-16B-137
24	87864	TL-72313
25	80006	AT-2B-137
25	87864	TL-72316
26	80006	AT-6B-137
26	87864	TL-72317
27	80006	AT-3B-137
27	87864	TL-72314
28	80006	AT-7B-137
28	87864	TL-72315
29	80006	AT-4B-137
30	80006	AT-8B-137
31	80006	AT-12B-112
31	87864	TL-68223
32	80006	AT-16B-112
32	87864	TL-68224
33	80006	AT-2B-112
33	87864	TL-68216
34	80006	AT-6B-112
34	87864	TL-68205

See footnote at end of table

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TABLE IV. Substitution data 1/ - continued

Dash number	Manufacturer's code (FSCM)	Manufacturer s part number
35	80006	AT-3B-112
35	87864	TL-68213
36	80006	AT-7B-112
36	87864	TL-68208
37	80006	AT-4B-112
38	80006	AT-8B-112
39	80006	AT-12B-090
39	87864	TL-64449
40	80006	AT-16B-090
40	87864	TL-64450
41	80006	AT-2B-090
41	87864	TL-64445
42	80006	AT-6B-090
42	87864	TL-64451
43	80006	AT-3B-090
43	87864	TL-64443
44	80006	AT-7B-090
44	87864	TL-64444
47	80006	AT-2B-062
47	87864	TL-57108
48	80006	AT-6B-062
48	87864	TL-57110
49	80006	AT-3B-062
49	87864	TL-57109
50	80006	AT-7B-062
50	87864	TFT-57067
51	80006	AT-4B-062
52	80006	AT-8B-062
53	80006	AT-2B-062

1/ Assemblies covered by this specification sheet are substitutable for the manufacturer's part numbers shown. This information in no way implies that the manufacturer's part is substitutable for the military part

Custodians
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Navy - EC
Air Force - 85

Review activities
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Navy - OS
Air Force - 17, 99
DLA - ES

User activities
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Navy - MC, CG, AS
Air Force - 11, 19

Preparing activity
Navy - EC

Agent
DLA - ES

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5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

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